

The Title of My M.Sc. Dissertation



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Acknowledgements

It doesn't really matter in which order the acknowledgements and abstract come. In some ways it's nice to put the acknowledgements first then everything that follows is about the project you have done.

You should generally thank your supervisor(s) in your acknowledgements — not doing so makes quite a statement. If anyone else has helped with the project then thank them too.

If you have received funding from anywhere, it is standard to thank your funding body.

It's then up to you who else to thank. You can be brief and stop there, or you can thank family, friends, the college cat . . .

Abstract

Your dissertation must contain an abstract, although it doesn't need to be very long. A paragraph or two is typical. However, if your abstract is long (i.e. it goes over one page) you might consider using the abstractlong environment instead, which has narrower margins.

The abstract should give a concise summary of what you will do in the dissertation.

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Chapter 1

Introduction

After the acknowledgements and abstract of your dissertation you should have a table of contents. If you wish you can also have a list of figures and a list of tables. Be aware though, that if you have long captions for the figures and tables, these lists can be very long and not necessarily very useful (you can also consider using truncated captions for the lists of figures and tables). The table of contents and the lists of figures and tables do not contribute towards the page count so you may wish to number these pages with Roman numbers.

1.1 Structure

The first chapter of your dissertation should be an introduction. If you used Roman numbers for the page numbering of the table of contents etc, you should start using standard Arabic numbers now. Ideally the first page of the introduction should be numbered page 1 as it is the first page that counts towards the page limit.

Think about sectioning your work nicely with sections, subsections and subsubsections.

1.1.1 Sectioning

Be consistent with capitalisation of titles of sections etc. Either capitalise all words (apart from of, the, a etc.) or just capitalise the first word and proper nouns. Choose one of these conventions and stick to it!

1.1.1.1 Numbering of Sections

By default sections and subsections will be numbered in the main body of the text and in the table of contents. You may wish to change this so that subsubsections are also numbered in the main text and perhaps in the table of contents. Commands of the form `\setcounter{secnumdepth}{3}` and `\setcounter{tocdepth}{3}` will do this for you.

1.2 About the Introduction

Use the introduction to explain what the problem is you are going to solve and how you will attack this (methods). You may wish to have a literature review in the introduction or in a separate chapter.

1.3 Main Contributions

The examiners have suggested that a “main contributions” section in the introduction is really useful, so please ensure you have such a section! You can use this to explain what you believe are the main results of your work. If any of what you have done is novel, it can be helpful to highlight that here. If you have worked on an industrial project you could state how and why your results are useful to your industrial partner.

1.4 Code

Here you may have a link to your GitHub repository (recommended but not mandatory). You can also make clear whether you have written code from scratch, or based it on other code, or had help from ChatGPT.

1.5 Structure of the Dissertation

It is common, although not compulsory, to end the introduction with a brief section explaining the structure of the dissertation. Often this goes along the lines of:

The remainder of the dissertation is structured as follows. In Chapter 2 we discuss blah blah. In Chapter 3 we blah blah. Finally in Chapter 4 we summarise the work we have done and suggest some areas for future research.

Chapter 2

The Work You've Done

After the introduction come chapters describing what you've actually done over the course of your research project.

I haven't done a dissertation so I will just insert a figure, see Figure 2.1, and a table, see Table 2.1, so that something appears in my lists of figures and tables. The figure caption also shows how you can give a shortened caption to appear in the list of figures. Make sure that you refer to all figures and tables from the main body of the text and make sure that all text which appears on figures is legible.

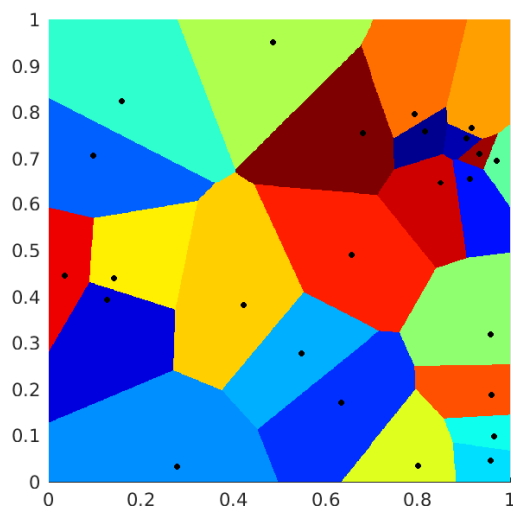


Figure 2.1: A Voronoi tessellation. The black dots represent 25 randomly chosen nodes in the unit square. The colouring at each other point indicates to which of the nodes it is closest.

The other thing you need to do is ensure you cite all the relevant materials you

mesh	<i>n_tri</i>	<i>n_nods</i>	estimated current	actual error	estimated error	effectivity index
1	32	25	1.270	0.270	0.829	3.07
2	94	59	1.131	0.131	0.413	3.16
3	201	116	1.066	0.066	0.215	3.23
4	372	208	1.034	0.034	0.115	3.34
5	527	288	1.019	0.019	0.069	3.57
6	720	391	1.011	0.011	0.042	3.81

Table 2.1: The triangulations produced by the adaptive finite element method for the steady state disc electrode problem.

have used. Everything that appears in the References should have been cited in the text (if you use BibTeX it's hard not to do this). There are a variety of types of work you might wish to cite, for example a book [1], a journal article [2], a thesis [3], work in a collection [4] or a conference proceedings [6], or perhaps a technical report [5]. In all cases be sure to give enough detail of the reference to allow the reader to find it. If you cite webpages be sure to list the date accessed in the references as in the entry for [7].

You might also like to insert a bullet point list. For this you can decrease the spaces between the bullet points using the `itemsep` command, see below:

- Item 1
- Item 2
- Item 3

It can be nice to end each chapter with a brief comment about what you have achieved in that chapter and linking to the next chapter. For example you could say that you have looked at the stability and convergence rates of a numerical scheme in this chapter. In the next chapter you will use the scheme to solve a given problem.

Chapter 3

Conclusions and Future Work

Here we summarise the work that has been presented in this dissertation and discuss possible areas for future work.

3.1 Summary

Give a summary of what has been done. You might do this chapter by chapter but be sure to highlight all the important points.

3.2 Future Work

You don't actually have to list further work, but most people do this so it seems unusual not to. Just think what you would do next if you had more time on this project.

3.3 Conclusion

You might like to give a final conclusion so the reader is left remembering what you have done, rather than what you would do if there wasn't a submission deadline.

3.4 Page Numbering

Remember that the page count stops after the final page of your final chapter so the last page of the conclusions should be at most 54 to avoid a penalty.

Appendix A

Use of Appendices

There is no need to have appendices unless you want to and you should remember that, while the examiners are required to read the main body of your dissertation, they are not required to read the appendices. This means that there should be nothing in an appendix which is vital to the reading of the dissertation.

However, you may still wish to use appendices, for example to list notation, to give more details of calculations or to provide extra results.

The order of the appendices and the references is down to personal choice, since neither contribute to the page count. However, having appendices before references can make it easier for the reader to quickly find the references section which can be very useful.

References

- [1] M. Abramowitz and I.A. Stegun, editors. *Handbook of Mathematical Functions*. Dover, New York, 1970.
- [2] R.C. Ackerberg, R.D. Patel, and S.K. Gupta. The heat/mass transfer to a finite strip at small Péclet numbers. *J. Fluid Mech.*, 86:49–65, 1977.
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- [5] G. Beckett, J.A. Mackenzie, and M.L. Robertson. A moving mesh finite element method for the solution of two-dimensional Stefan problems. Technical Report 99/26, Department of Mathematics, University of Strathclyde, Livingston Tower, 26 Richmond Street, Glasgow, G1 1XH, 1999.
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- [7] W. Strunk Jr. and E. B. White. *The Elements of Style*. Longman, fourth edition edition, 1999. <http://www.bartleby.com/141/> [accessed: 05-06-2024].